

Introduction to Artificial Intelligence

A short reference document for testing a Retrieval-Augmented Generation (RAG) system.

1. What is Artificial Intelligence?

Artificial Intelligence, commonly called AI, is the field of computer science concerned with building systems that can perform tasks that normally require human intelligence. These tasks can include learning from data, recognizing patterns, understanding language, making predictions, solving problems, and making decisions.

Modern AI systems often rely on machine learning, where a computer learns patterns from examples instead of being explicitly programmed with every rule. Deep learning is a subset of machine learning that uses neural networks with many layers to learn complex representations from large datasets.

2. Machine Learning

Machine learning is an approach in which algorithms learn relationships and patterns from data. The three commonly discussed categories are supervised learning, unsupervised learning, and reinforcement learning.

Supervised Learning

In supervised learning, a model is trained using examples that contain both input data and known target outputs. Classification predicts discrete categories, such as whether an email is spam. Regression predicts continuous values, such as the expected price of a house.

Unsupervised Learning

Unsupervised learning works with data that does not have labeled target values. Clustering is a common example. A clustering algorithm can group similar observations together based on their characteristics.

Reinforcement Learning

Reinforcement learning involves an agent interacting with an environment. The agent receives rewards or penalties and learns a strategy that aims to maximize its long-term reward.

3. Neural Networks

A neural network is a machine learning model made of interconnected computational units called neurons. Neurons are commonly organized into an input layer, one or more hidden layers, and an output layer.

During training, the network adjusts its parameters so that its predictions become closer to the desired outputs. A process called backpropagation calculates how the parameters contributed to the prediction error, while an optimization algorithm updates them.

Deep neural networks contain multiple hidden layers. They are particularly useful for tasks involving images, audio, natural language, and other high-dimensional data.

4. Natural Language Processing

Natural Language Processing (NLP) focuses on enabling computers to work with human language. Applications include text classification, sentiment analysis, machine translation, question answering, summarization, and conversational assistants.

A major challenge in NLP is that the meaning of a word or sentence depends heavily on context. Modern language models address this by learning relationships between words and larger pieces of text.

5. Large Language Models

Large Language Models (LLMs) are neural networks trained on large collections of text. They learn statistical patterns in language and can generate text based on an input prompt. LLMs can be used for tasks such as question answering, summarization, code generation, and text transformation.

However, an LLM does not automatically have access to a private document or a company's internal knowledge. Retrieval-Augmented Generation can address this limitation by retrieving relevant external information and providing it to the model as context.

6. Retrieval-Augmented Generation

Retrieval-Augmented Generation, or RAG, is an architecture that combines information retrieval with text generation. A collection of documents is first processed into smaller chunks. Each chunk is converted into an embedding vector and stored in a vector database.

When a user asks a question, the question is also converted into an embedding. A similarity search finds the document chunks whose meanings are closest to the question. These retrieved chunks are then supplied to an LLM along with the question.

RAG pipeline: Document → Chunking → Embeddings → Vector Database → Similarity Search → Retrieved Context → LLM → Answer

7. Embeddings

An embedding is a numerical representation of text. Instead of representing a sentence only as individual words, an embedding model maps the sentence into a vector containing many numerical dimensions. Sentences with similar meanings tend to have similar vector representations.

For example, the questions 'What is AI?' and 'Explain artificial intelligence' are written differently but have similar meanings. A good embedding model should place their vectors relatively close together.

8. Vector Databases

A vector database stores embedding vectors and supports similarity search. Examples include Chroma, Qdrant, and PostgreSQL with the pgvector extension. Persistent vector databases allow embeddings to remain available after the application stops.

A vector database is useful in RAG because searching vectors by semantic similarity is different from a traditional keyword search. The retrieved text can then be used as evidence or context for an LLM.

9. Example Questions for a RAG System

This document can be used to test retrieval with questions such as: What is artificial intelligence? What are the three common types of machine learning? What is supervised learning? What is reinforcement learning? What is an embedding? Why are vector databases useful in RAG? What is the purpose of retrieval in a RAG pipeline?